

Review

Carbon allotropes: beyond graphite and diamond

Eduardo HL Falcao* and Fred Wudl*

University of California, Santa Barbara, USA

Abstract: Carbon offers unmatched versatility among the elements of the periodic table. Depending on its hybridization state and atomic arrangement, carbon forms the layered semiconductor graphite, the insulator diamond, with its unmatched hardness, the high surface area amorphous carbons, and the nano-sized fullerenes and nanotubes, among others. This paper broadly reviews carbon allotropes, with emphasis on some recent results and applications, particularly in the field of high surface area amorphous carbons.

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Keywords: carbon; amorphous carbon; fullerenes; surface area; adsorption; chemically synthesized amorphous carbons

INTRODUCTION

Over 95% of all known chemical compounds are 'carbon compounds'. Although some other elements can bond to themselves to form chains and clusters, none of them equals carbon in the extent and versatility of its catenation. Carbon bonds to both electronegative and electropositive elements and, more importantly, it can bond to itself essentially without limits through single (average bond energy $\sim 350 \text{ kJ mol}^{-1}$), double ($\sim 610 \text{ kJ mol}^{-1}$), or triple ($\sim 840 \text{ kJ mol}^{-1}$) bonds.¹ High molecular weight polymers, diamond and graphite provide illustrative examples of such catenating power. The latter two are composed exclusively of carbon atoms and have been traditionally considered the main allotropic forms of the element. Amorphous carbons consist of nearly pure carbon with varying $\text{sp}^2\text{:sp}^3$ ratios, and have been considered as another allotropic form of the element (although lacking any crystallinity and often containing significant amounts of hydrogen). Other allotropes of carbon include fullerenes, nanotubes and the controversial carbynes. The diagram in Fig. 1 summarizes these main known allotropic forms of carbon.² Table 1 lists the allotropes, selected properties and applications, as well as their availability (or current synthesis scale status).

DIAMOND, GRAPHITE, CARBYNE

Diamond is the hardest substance known (10 in the Mohs hardness scale), and it is the thermodynamically stable form of carbon above 60 kbar.³ It is a precious gem and an important abrasive material. It usually crystallizes with a cubic structure, and its industrial

synthesis has been realized.^{4,5} It also exists in the hexagonal form, less common in nature and with lower hardness (as the mineral Lonsdaleite, 7–8 in the Mohs scale). The two forms can be interconverted under specific conditions.⁶ In diamond, each sp^3 -hybridized carbon atom covalently bonds to four others, extending three-dimensionally as an array of tetrahedra. Diamond transforms into graphite, the thermodynamically stable allotrope at lower pressures, above 1500°C under vacuum or inert atmosphere.³ Under ambient conditions such interconversion is negligibly slow.

Ideally, graphite comprises infinite layers of sp^2 -hybridized carbon atoms. Within a layer (called a graphene sheet), each C atom bonds to three others, forming a planar array of fused hexagons. The un-hybridized $2p_z$ orbital that accommodates the fourth electron forms a delocalized orbital of π symmetry that further stabilizes the in-plane bonds. The weaker van der Waals force keeps the layers stacked together, $3.354 \text{ \AA}$ apart.⁷ Hexagonal graphite, the most common form, stacks in an ABAB fashion. An ABCABC stacking characterizes the rarer rhombohedral graphite. Graphite's lubricant properties originate from its layered structure and from the van der Waals force between the layers that allow the layers to slide past each other. Graphite is also a good electrical and thermal conductor in the plane directions (due to the delocalized π band and the σ bonds, respectively), and a poor electrical and thermal conductor in the stacking direction (due to the van der Waals force between the layers).⁷ The electrical conductivity of graphite at room temperature in the basal (a-direction) and prismatic (c-direction)

* Correspondence to: Eduardo HL Falcao, Materials Research Laboratory, University of California, Santa Barbara, CA 93106, USA and Fred Wudl, Department of Chemistry and Biochemistry, University of California, Santa Barbara, CA 93106, USA

E-mail: efalcao@mrl.ucsb.edu, wudl@chem.ucsb.edu

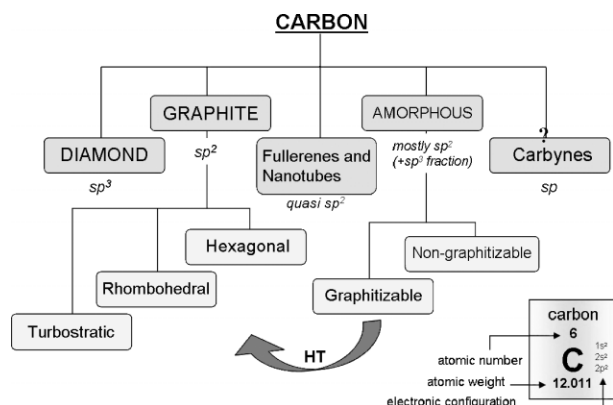
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Table 1. Carbon allotropes, availability, selected properties and applications

Substance	Availability	Selected properties	Selected applications
Diamond	Naturally occurring, artificial production well established; commercially available.	Hardness, wide bandgap, thermal conductivity.	Abrasives, jewelry.
Graphite	Naturally occurring, artificial production well established; commercially available.	Anisotropic electronic conductivity, lubricity.	Lubricants, electrode component.
Carbyne	Low stability; detected in space, oligomers synthesized in laboratory scale.	Under investigation.	Under investigation (mostly: oligomers).
Fullerenes (esp. C ₆₀)	Large laboratory scale synthesis, industrial production in development stages; commercially available.	Electronic and optical properties, potential high hardness in composites, superconducting upon doping.	Hybrid solar cells. Proposed: nanoelectronics, pharmaceuticals etc. (in general: derivatives).
Nanotubes	Laboratory scale synthesis, structural control and purification still in development; commercially available.	Electronic properties; high strength.	Proposed: structural reinforcement, nanoelectronics, pharmaceuticals.
Carbon onions	Laboratory scale synthesis.	Under investigation.	Under investigation. Proposed: nanoscale protective shells and reactors.
Graphene films	Laboratory scale synthesis.	Unique electronic properties.	Under investigation. Proposed: nanoelectronics.
Amorphous carbons (mostly sp ²)	Large industrial scale synthesis, commercially available.	High surface area (~500–3000 m ² g ⁻¹).	Adsorbents, catalysis, electrodes in batteries.
Amorphous carbons (mostly sp ³)	Films prepared <i>in situ</i> for specific applications, well established.	High hardness, electrically insulating.	Electronics.
Carbon foams	Laboratory scale synthesis.	Relatively high surface area; unique magnetic properties.	Under investigation.
Carbon fibers	Industrial production well established; commercially available.	Electrical and thermal conductivity, high strength.	Structural reinforcer in several applications.
Chemically synthesized amorphous carbons (incl. carbide-derived carbons and carbons prepared from halogenated precursors)	Laboratory scale.	High surface area; for CDCs, pore size tunability.	Proposed: adsorption, catalysis, energy storage.

planes are, respectively, $\sigma_{a(RT)} \sim 2-3 \times 10^4 \text{ S cm}^{-1}$ and $\sigma_{c(RT)} \sim 6 \text{ S cm}^{-1}$.^{7,8} The thermal conductivity values are $K_{a(RT)} \sim 15-20 \text{ W cm}^{-1} \text{ K}^{-1}$ and $K_{c(RT)} \sim 0.05-0.1 \text{ W cm}^{-1} \text{ K}^{-1}$.^{9,10} It must be noted that these values strongly depend on the sample history and temperature.

**Figure 1.** Allotropes of the element carbon.²

Relative rotations or translations of adjacent layers give rise to stacking faults. In this case, the regular stacking patterns are lost and the interlayer spacing becomes variable and slightly larger than in graphite. Turbostratic graphite has this structure, and it generally does not convert into graphite with high temperatures alone (although a progressive decrease in the interlayer spacing does take place).¹¹ Pyrolytic graphite is a polycrystalline artificial graphite prepared by pyrolysis of a carbonaceous gas above 2000 °C. Although well ordered in the stacking direction, the graphene planes are more or less randomly oriented. Annealing treatment produces highly oriented pyrolytic graphite (HOPG), which has a mosaic spread (plane misalignment) of $\sim 0.02^\circ$ and crystallite sizes of $\sim 1 \mu\text{m}$.⁷

The existence of materials containing purely sp-hybridized carbon atoms in the form of linear chains has been the subject of controversy for a long time.¹²⁻¹⁵ It has been hypothesized that interstellar clouds contain significant amounts of such

compounds.^{12,16} The chains can be composed either by alternating triple and single bonds or by cumulated double bonds (polynes and polycumulenes, respectively), both of which are, expectedly, highly reactive and thermally unstable.^{17,18} The study of the elemental solid composed exclusively of sp-hybridized carbon, called carbyne, is hence very limited, although great progress in the synthesis of end-capped oligynes and short polynes has been made recently.^{19–25} These molecules are attractive due to their interesting properties and possible applications as molecular wires. Carbyne itself, on the other hand, still remains mostly a subject of academic and fundamental interest, due to the difficulty in preparation and instability.²⁵

FULLERENES, NANOTUBES, CARBON ONIONS AND OTHER NANOSTRUCTURES

Recently, other allotropic forms of carbon have been reported, such as fullerenes (including C₆₀ and C₇₀)^{26–33} and nanotubes.^{30,34–37} Fullerenes and single-walled carbon nanotubes (SWNTs) form when the dangling bonds at the edges of a real (finite) graphene layer are connected together. The total energy of the layer is therefore reduced, although bending the graphene layer increases the strain energy (similar to what happens when replacing some hexagon rings by pentagons in order to form a spherical structure), which causes the hybridization state to no longer be purely sp².^{29,30,32,35} The chemistry and physical properties of fullerenes have received much attention in recent years, with several functionalized derivatives being reported.^{28,31,33} Solar cell³⁸ and biological applications^{39,40} based on the many interesting physical properties^{29,41} of C₆₀ and its derivatives have been recently reviewed. The C₆₀ derivative (6,6)-phenyl-C₆₁-butyric acid methyl ester (PCBM)⁴² is currently used as the electron acceptor in commercial organic solar cells.^{33,38,43–45}

Carbon nanotubes have also been the subject of increased research interest, especially because of their electronic and mechanical properties. Depending on the way the graphene sheet was rolled to form the nanotube cylinder, a metallic or semiconducting behavior arises.^{35,37} Nanotubes can also have different diameters, be open-ended or closed-ended (in which case they will resemble a very long fullerene), and be single-walled or multi-walled (in which case a concentric array of carbon cylinders ~3.4 Å apart is observed), and either of these parameters will affect the material's properties (although fine control of the end product is an exceedingly difficult endeavor). Nanotubes and chemically modified nanotubes have the potential for such diverse application as nanoelectronics/nanospintronics components,^{46–49} reinforcements in composite materials,⁵⁰ and several biological applications, including biosensing, tissue engineering and drug-delivery.^{51–54} Carbon nanotubes have also been

studied for applications as a hydrogen storage medium. Later work⁵⁵ has cast doubts on the initial reports of very high H₂ storage capacities. More recently, however, some careful studies have reported storage capacities up to 3–4 wt% (under specific conditions), which could generate renewed interest in this field.^{56–59} Amorphous porous carbons, however, still outperform nanotubes in terms of physisorptive storage capacity, and may be obtained at significantly lower costs.

Carbon onions can be considered as multi-walled fullerenes.^{30,60–62} These remarkably spherical concentric shells were observed when soot was exposed to the intense electron beam of a high-resolution electron microscope.⁶⁰ Other strategies for forming carbon onions ensued, such as the use of arc-discharge,^{61,63,64} the thermal treatment of soot,⁶⁵ or chemical vapor deposition of carbon over a nickel/aluminium catalyst.⁶⁶ As with fullerenes and nanotubes, studies of the functionalization and properties of this material soon followed.^{67,68} An interesting potential application for the carbon onions involve encapsulating another material, most notably metals.^{69–72} The carbon shell could then work as a shield in order to protect the metal against oxidation.⁷⁰ The system can also be used as a “nano-reactor”.⁷² In this case, electron beam irradiation of carbon onions containing nanometer-size iron crystals led to the formation of cementite (iron carbide), as followed by the lattice evolution (disappearance of Fe fringes and subsequent appearance of fringes corresponding to Fe₃C). Repetition of the experiment with encapsulated cobalt crystals led only to Co diffusion and shrinkage of the particles, while the Co lattice remained unchanged during the whole irradiation time.⁷²

It has recently been reported that the complete detachment of graphene sheets leads to the formation of carbon nanoscrolls with diameters of few tens of nanometers.^{73,74} The individual graphene sheet rolls around itself in order to minimize the surface energy.⁷⁴ The newly formed curved layered structure resembles that of a multi-walled nanotube, with a spacing of ~3.4 Å.⁷³ Similar structures, with diameters ranging from fractions of a micrometer up to ~5 μm and lengths of up to 3 cm, have been observed in an arc discharge. These microscrolls, named carbon whiskers, behave essentially like single crystals and have high flexibility and strength.⁷⁵

Another very recent development closely related to fullerenes and nanoscrolls has been called atomically thin carbon films. Films containing a few graphene layers (down to single-layer) were produced by repeated mechanical exfoliation of HOPG,⁷⁶ and showed electric field⁷⁶ and quantum Hall⁷⁷ effects, which may lead to applications in nanoelectronic devices. Later theoretical studies aimed at explaining those results,^{78,79} and other techniques to obtain and characterize these ultra-thin films continue to attract interest.⁸⁰

AMORPHOUS CARBONS I

Before the advent of carbynes, fullerenes and nanotubes, amorphous carbons (or “non-graphitic” carbons) were considered as the third allotropic form of carbon.⁸¹ These materials feature roughly planar layers of mostly sp^2 -hybridized carbon atoms but they lack long-range crystallinity, especially in the stacking direction. They contain a significant fraction of sp^3 carbon atoms, which often crosslink neighboring layers, forming an overall structure that comprises amorphous and more graphitic regions.^{82,83} Amorphous carbons are usually prepared by pyrolysis of organic polymer or hydrocarbon precursors at temperatures below 1500 °C. The nature of the starting material and the preparation history strongly influence the structural development and final properties upon further heat treatment.⁸³ Some amorphous carbons also contain an expressive amount of hydrogen, mostly bound on the surface of the carbon particles, which would obviously preclude the application of the term “allotrope” to such materials.

The so-called graphitizable carbons develop a graphitic structure continuously with heating to temperatures ranging from 1500 to 3000 °C, because the degree of crosslinking between the ordered sp^2 regions is very low, and favorable orientation easily takes place. Materials prepared from liquid precursors, such as petroleum pitch, show higher crystallinity or at least favor graphitization (e.g. cokes, artificial graphites). Graphitizable carbons are also called “soft carbons”.^{11,83} Materials prepared from the gas phase can have a very high degree of three-dimensional order upon prolonged thermal treatment (e.g. vapor-grown carbon fibers, pyrolytic graphite).¹¹ Carbon blacks are usually prepared from acetylene, natural gas or coal tar distillates, and show a high degree of crystallinity, with a small fraction of amorphous regions blended within the particles.⁸⁴

Non-graphitizable carbons (e.g. chars and glassy carbons), also called “hard carbons”, originate from solid precursors, such as polymers and resins and have a highly disordered structure.^{11,82,83} These carbons are isotropic in all properties, and do not develop significant graphitic structure even at high temperatures (2500–3000 °C), mostly due to a high degree of crosslinking that locks the crystalline regions.

Activated carbons are amorphous (usually hard) carbons that underwent chemical or physical activation.⁸² They are also generically called “porous carbons”. The activation process gives rise to very high surface areas ($\sim 500\text{--}3000\text{ m}^2\text{ g}^{-1}$) and sometimes to a well-developed microporosity. Chemical activation involves the addition of compounds such as ZnCl_2 , H_3PO_4 and H_2SO_4 to the precursor feedstock prior to carbonization, and later removal. This process is currently not favored because of environmental considerations. Physical activation refers to treatment of the carbon material by gases such as carbon dioxide and steam in the temperature range 700–1100 °C. Industrially, activated carbons are produced by the

carbonization and physical activation of coal, lignite, wood and coconut shell, for example.^{7,82} Activated carbons have been known since ancient times, and have been used in water treatment, in gas separation, as catalysts and catalyst supports, as filter materials and as decolorizing agents.^{7,82,84} They have also been studied for the physisorptive storage of hydrogen, with some encouraging results due to the high surface areas attainable.⁸⁵

Carbon fibers are an important class of carbon materials, also closely related to multi-walled carbon nanotubes,⁸¹ and having a carbon content higher than 92%.⁷ Their preparation resembles that of amorphous carbon, usually making use of pyrolysis of an organic precursor under specific conditions. Although ultimately the orientation of the graphene planes depends on the precursors used, the growth of bent sheets parallel to the fiber axis seems to be preferred.³⁵ Moreover, depending on the synthesis conditions, the fibers can be amorphous or exhibit different degrees of crystallinity (up to nearly 100%), which will affect the fiber’s ultimate properties and target applications.⁷

Carbon aerogels also form from the pyrolysis of organic precursors, in this case organic aerogels produced through sol–gel processes (such as resorcinol-formaldehyde sol–gels).^{86,87} These materials usually have low density and very high specific surface areas (up to $\sim 1200\text{ m}^2\text{ g}^{-1}$), and they can be produced in different forms, such as monoliths, fine particles or films. The final shape and properties depend strongly on the sample history,^{86–89} as is the case with all amorphous carbons. Carbon aerogels are promising materials for use as adsorbents and catalysts,^{86,89} as well as electrodes in double-layer capacitors.⁹⁰

Amorphous diamond-like carbons (DLCs) are amorphous carbons containing a very high fraction of sp^3 carbon atoms, sometimes as high as 85%.^{91–95} DLCs are sometimes called highly tetrahedral amorphous carbons, or ta-C, and they resemble amorphous silicon in several respects, which has enabled some electronic applications. Depending on the sample history, the material may be hydrogen-free or hydrogenated. They are usually formed as thin films under specific deposition conditions, such as ion beam deposition, laser ablation or plasma enhanced chemical vapor deposition, the latter being the usual method for producing the hydrogenated carbon films.^{92,94,95} Due to the material’s hardness, films of DLC have been used as protective coatings for hard disks and optical parts.

A low-density carbon nanofoam with a significant ($\sim 35\%$) fraction of sp^3 carbons has been produced by ultrafast ablation with a high pulse-rate Nd-YAG laser.⁹⁶ The nanofoam forms as a granular material, with an estimated specific surface area of about $300\text{ m}^2\text{ g}^{-1}$. It also shows an unusually high concentration of unpaired spins that results in a paramagnetic susceptibility, in contrast to the

characteristic diamagnetism of most other allotropes of carbon.^{97,98}

AMORPHOUS CARBONS II – THE CHEMICAL APPROACH

The amorphous carbons reviewed previously either form from the pyrolysis of organic precursors or deposit as a film from the gas generated by a physical process such as laser ablation or ion bombardment. Two other approaches rely on chemical reactions between specific starting materials to produce carbons with interesting morphologies and/or tunable properties. The chlorination of metal carbides (including silicon and boron carbides) affords the so-called carbide-derived carbons.^{99–103} The de-halogenation (most notably the de-chlorination) of small organic molecules leads to the formation of carbonaceous clusters with very interesting morphologies.^{104–106}

Carbide-derived carbons (CDCs) form when chlorine gas extracts the metal from carbides such as Ti_3SiC_2 , SiC , B_4C or ZrC , leaving behind a framework of nearly pure carbon. The rigid carbide lattice, which acts as a template, and the progressive layer-by-layer extraction result in a material with a very narrow pore size distribution, comparable to that of zeolites.¹⁰⁰ Careful selection of the carbide precursor and of the reaction temperature (which usually ranges from 200 to 1200 °C) enables fine-tuning the CDC porosity. These materials can be obtained as a powder, film, or bulk pieces.^{100–102,107} CDCs are promising materials for applications in hydrogen storage,^{107–109} gas separation,¹⁰⁰ electrochemical capacitors,^{108,110} and removal of inflammatory proteins.¹¹¹

The Wurtz-type reaction of halogenated compounds with an alkali metal has been used to produce carbon materials.^{104–106} Decomposition of ferrocene ($\text{Fe}(\text{C}_5\text{H}_5)_2$) in the presence of benzene or tetrachloroethylene^{112,113} and pyrolysis of hexabromocyclopentadiene¹¹⁴ have also been reported. This approach succeeded in preparing carbon microrods¹⁰⁶ and other carbon micro- and nanostructures.^{104,105,112,113}

The carbon microrods were prepared by reaction between hexachlorobenzene (C_6Cl_6) and lithium metal at 250 °C in a sealed tube. The rod-like morphology is the major product of the

reaction, with disordered layers of mostly sp^2 carbons stacked perpendicular to the rod long axis.¹⁰⁶ This material has relatively high surface areas (200 to 340 $\text{m}^2 \text{g}^{-1}$, depending on heat treatment) and shows a reversible lithium storage capacity in excess of 500 mA h g^{-1} .^{115,116} Although a large irreversible capacity and moderate efficiency may limit application as anode in Li-ion batteries, their unique shape and dimensions make them promising candidates for three-dimensional battery architectures.^{117–119} An SEM image of the carbon microrods is shown in Fig. 2(a).

In a similar reaction, sodium metal reacted with hexachlorobenzene at 160 °C¹⁰⁵ or with hexachlorocyclopentadiene (Cl_6C_5 or Cl_6Cp) at 100–150 °C¹⁰⁴ in sealed vessels. Spherical or elliptical hollow particles with varying sizes resulted from these reactions. Since Cl_6Cp is a liquid at room temperature and up to ~ 239 °C, an alternative procedure used the slow addition of lithium sand to hot Cl_6Cp (Falcao *et al.*, in prep.). This approach afforded a carbon material with a different morphology, one that resembles that of traditional amorphous carbon, as seen in Fig. 2(b). The material has a very high specific surface area, up to 3200 $\text{m}^2 \text{g}^{-1}$ after activation at 1000 °C under an argon flow. The fact that this material can be activated under an inert atmosphere contrasts with most amorphous carbons, which require chemical (acid) or physical (CO_2 or steam) activation in order to develop their ultimate porosity.⁸² Expectedly, this material outperformed activated carbon (surface area $\sim 1000 \text{ m}^2 \text{g}^{-1}$) in preliminary tests in terms of hydrogen adsorption capacity (Falcao *et al.*, in prep.), as illustrated in Fig. 3.

SUMMARY

Carbon offers unmatched versatility among the elements of the periodic table. Its best known allotropes, graphite and diamond, illustrate the range of properties it can display: while one is a black, soft material with conducting and lubricant properties, the other is an extremely hard, transparent insulator. Fullerenes and nanotubes appeared in a relatively recent timeframe, but their impact in science is striking, as numerous groups continue to investigate their properties and applications. Other even more recent structures, such as onions, atomically thin

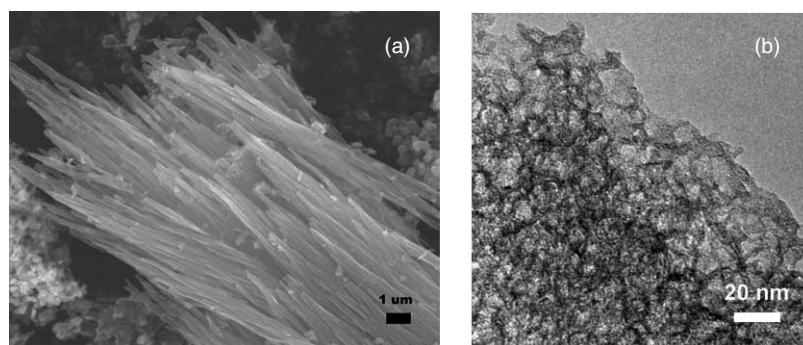


Figure 2. (a) SEM image of carbon microrods; (b) TEM image of a porous carbon sample synthesized from Cl_6Cp .

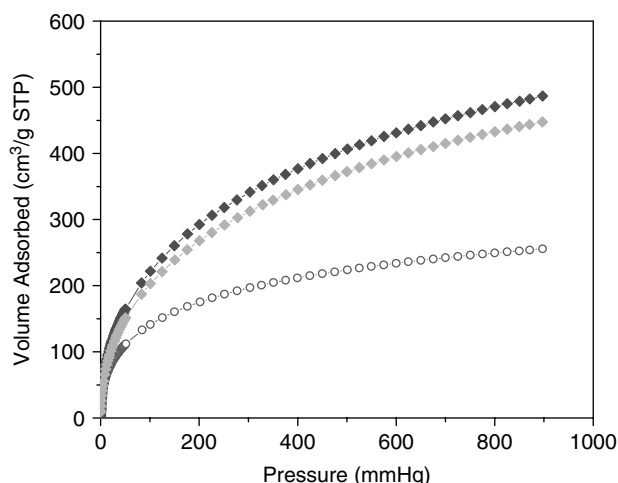


Figure 3. Hydrogen adsorption of a highly porous carbon sample synthesized from Cl_6Cp (first adsorption, dark diamonds; second adsorption, light diamonds) and activated carbon (open circles).

films and nanofoams, further illustrate carbon's rich chemistry and potential. Amorphous carbons have been known and used since ancient times and recent results on alternative preparation methods show that there is still room for improvements and discoveries, particularly in the control of pore sizes.

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REFERENCES

- Streitwieser A, Heathcock CH and Kosower EM, *Introduction to Organic Chemistry*, 4th Edition, Macmillan Publishing Company, New York, Ch. 1 and 2 (1992).
- Falcao EHL, Carbonaceous materials with exotic morphologies. PhD Dissertation, University of California, Los Angeles, CA, Ch. 1 (2006).
- Mantell CL, *Carbon and Graphite Handbook*, Interscience Publishers, New York, Ch. 3 (1968).
- Devries RC, Synthesis of diamond under metastable conditions. *Ann Rev Mater Sci* 17:161–187 (1987).
- Ferro S, Synthesis of diamond. *J Mater Chem* 12:2843–2855 (2002).
- He H, Sekine T and Kobayashi T, Direct transformation of cubic diamond to hexagonal diamond. *Appl Phys Lett* 81:610–612 (2002).
- Chung DDL, Graphite. *J Mater Sci* 37:1475–1489 (2002).
- Spain IL, Ubbelohde AR and Young DA, Electronic properties of well oriented graphite. *Philos Trans Royal Soc A* 262:345–386 (1967).
- Hooker CN, Ubbelohde AR and Young DA, Anisotropy of thermal conductance in near-ideal graphite. *Proc Royal Soc A* 284:17–31 (1965).
- Taylor R, Gilchrist KE and Poston LJ, Thermal conductivity of polycrystalline graphite. *Carbon* 6:537–544 (1968).
- Imanishi N, Takeda Y, Yamamoto O, in *Lithium Ion Batteries – Fundamentals and Performance*, ed. by Wakihara M and Yamamoto O, Wiley-VCH, Weinheim, Germany, Ch. 5 (1998).
- Whittaker AG, Watts EJ, Lewis RS and Anders E, Carbynes – Carriers of primordial noble-gases in meteorite. *Science* 209:1512–1514 (1980).
- Jansta J and Dousek FP, Some aspects of existence of elementary carbon with sp-hybridized bonds. *Carbon* 18:433–437 (1980).
- Smith PPK and Buseck PR, Carbyne forms of carbon: Do they exist? *Science* 216:984–986 (1982).
- Heimann RB, Kleiman J and Salansky NM, Structural aspects and conformation of linear carbon polytypes (carbynes). *Carbon* 22:147–156 (1984).
- Henning T and Salama F, Carbon in the universe. *Science* 282:2204–2210 (1998).
- Casari CS, Li Bassi A, Ravagnan L, Siviero F, Lenardi C, Piseri P, *et al*, Chemical and thermal stability of carbyne-like structures in cluster-assembled carbon films. *Phys Rev B* 69:754221–754227 (2004).
- Yamada K, Tanabe Y and Sawaoka AB, Allotropes of carbon shock synthesized at pressures up to 15GPa. *Philos Mag A* 80:1811–1828 (2000).
- Hlavaty J, Kavan L, Kasahara N and Oya A, Polymerisation of 1-iodohexa-1,3,5-triyne and hexa-1,3,5-triyne: a new synthesis of carbon nanotubes at low temperatures. *Chem Commun* 737–738 (2000).
- Mohr W, Stahl J, Hampel F and Gladysz JA, Bent and stretched but not yet the breaking point: $\text{C}_8\text{--C}_{16}$ sp carbon chains that span two platinum atoms and the first structurally characterized 1,3,5,7,9,11,13,15-octayne. *Inorg Chem* 40:3263–3264 (2001).
- Kavan L, Nanocarbons made by soft chemistry. *Mol Cryst Liq Cryst* 386:167–172 (2002).
- Gibtnier T, Hampel F, Gisselbrecht JP and Hirsch A, End-cap stabilized oligoynes: Model compounds for the linear sp carbon allotrope carbyne. *Chem Eur J* 8:408–432 (2002).
- Cataldo F, Polyynes: a new class of carbon allotropes. About the formation of dicyanopolynes from an electric arc between graphite electrodes in liquid nitrogen. *Polyhedron* 23:1889–1896 (2004).
- Eisler S, Slepov AD, Elliott E, Luu T, McDonald R, Hegmann FA, *et al*, Polyynes as a model for carbyne: synthesis, physical properties, and nonlinear optical response. *J Am Chem Soc* 127:2666–2676 (2005).
- Cataldo F, A method for synthesizing polyynes in solution. *Carbon* 43:2792–2800 (2005).
- Kroto HW, Heath JR, O'Brien SC, Curl RF and Smalley RE, C_{60} : Buckminsterfullerene. *Nature* 318:162–163 (1985).
- Kroto HW, Allaf AW and Balm SP, C_{60} : Buckminsterfullerene. *Chem Rev* 91:1213–1235 (1991).
- Wudl F, The chemical properties of buckminsterfullerene (C_{60}) and the birth and infancy of fullerenes. *Acc Chem Res* 25:157–161 (1992).
- Dresselhaus MS, Dresselhaus G and Eklund PC, Fullerenes. *J Mater Res* 8:2054–2097 (1993).
- Rao CNR, Seshadri R, Govindaraj A and Sen R, Fullerenes, nanotubes, onions and related carbon structures. *Mater Sci Eng R* 15:209–262 (1995).
- Thilgen C, Herrmann A and Diederich F, The covalent chemistry of higher fullerenes: C_{70} and beyond. *Angew Chem Int Ed Engl* 36:2268–2280 (1997).
- Buhl M and Hirsch A, Spherical aromaticity of fullerenes. *Chem Rev* 101:1153–1183 (2001).
- Wudl F, Fullerene materials. *J Mater Chem* 12:1959–1963 (2002).
- Iijima S, Helical microtubules of graphitic carbon. *Nature* 354:56–58 (1991).
- Dresselhaus MS and Endo M, Relation of carbon nanotubes to other carbon materials. *Topic Appl Phys* 80:11–28 (2001).

- 36 Andrews R, Jacques D, Qian DL and Rantell T, Multiwall carbon nanotubes: Synthesis and application. *Acc Chem Res* **35**:1008–1017 (2002).
- 37 Dai H, Carbon nanotubes: Synthesis, integration and properties. *Acc Chem Res* **35**:1035–1044 (2002).
- 38 Segura JL, Martin N and Guldi DM, Materials for organic solar cells: the C_{60}/π -conjugated approach. *Chem Soc Rev* **34**:31–47 (2005).
- 39 Jensen AW, Wilson SR and Schuster DI, Biological applications of fullerenes. *Bioorg Med Chem* **4**:767–779 (1996).
- 40 Bosi S, Da Ros T, Spalluto G and Prato M, Fullerene derivatives: an attractive tool for biological applications. *Eur J Med Chem* **38**:913–923 (2003).
- 41 Foote CS, Photophysical and photochemical properties of fullerenes. *Topic Curr Chem* **169**:347–363 (1994).
- 42 Hummelen JC, Knight BW, LePeq F, Wudl F, Yao J and Wilkins CL, Preparation and characterization of fulleroid and methanofullerene derivatives. *J Org Chem* **60**:532–538 (1995).
- 43 Zheng L, Zhou Q, Deng X, Sun D, Yuan M, Yu G, *et al*, Synthesis of C_{60} derivatives for polymer photovoltaic cell. *Synth Met* **135–136**:827–828 (2003).
- 44 Hoppe H, Egbe DAM, Muehlbacher D and Sariciftci NS, Photovoltaic action of conjugated polymer/fullerene bulk heterojunction solar cells using novel PPE–PPV copolymers. *J Mater Chem* **14**:3462–3467 (2004).
- 45 Thompson BC, Kim YG, McCarley TD and Reynolds JR, Soluble narrowband gap and blue propylenedioxithiophene–cyanovinylene polymers as multifunctional materials for photovoltaic and electronic applications. *J Am Chem Soc* **128**:12714–12725 (2006).
- 46 Avouris P, Molecular electronics with carbon nanotubes. *Acc Chem Res* **35**:1026–1034 (2002).
- 47 Zhao JJ and Xie RH, Electronic and photonic properties of doped carbon nanotubes. *J Nanosci Nanotechnol* **3**:459–478 (2003).
- 48 Terrones M, Carbon nanotubes: synthesis and properties, electronic devices and other emerging applications. *Int Mater Rev* **49**:325–377 (2004).
- 49 Cottet A, Kontos T, Sahoo S, Man HT, Choi MS, Belzig W, *et al*, Nanospintronics with carbon nanotubes. *Semicond Sci Technol* **21**:S78–S95 (2006).
- 50 Coleman JN, Khan U and Gun'ko YK, Mechanical reinforcement of polymers using carbon nanotubes. *Adv Mater* **18**:689–706 (2006).
- 51 Lin Y, Taylor S, Li HP, Fernando KAS, Qu LW, Wang W, *et al*, Advances toward bioapplications of carbon nanotubes. *J Mater Chem* **14**:527–541 (2004).
- 52 Polizu S, Savadogo O, Poulin P and Yahia L, Applications of carbon nanotubes–based biomaterials in biomedical nanotechnology. *J Nanosci Nanotechnol* **6**:1883–1904 (2006).
- 53 Harrison BS and Atala A, Carbon nanotube applications for tissue engineering. *Biomaterials* **28**:344–353 (2007).
- 54 Bianco A, Kostarelos K and Prato M, Applications of carbon nanotubes in drug delivery. *Curr Opin Chem Biol* **9**:674–679 (2005).
- 55 Tibbetts GG, Meisner GP and Olk CH, Hydrogen storage capacity of carbon nanotubes, filaments, and vapor-grown fibers. *Carbon* **39**:2291–2301 (2001).
- 56 Gundiah G, Govindaraj A, Rajalakshmi N, Dhathathreyan KS and Rao CNR, Hydrogen storage in carbon nanotubes and related materials. *J Mater Chem* **13**:209–213 (2003).
- 57 Takagi H, Hatori H, Soneda Y, Yoshizawa N and Yamada Y, Adsorptive hydrogen storage in carbon and porous materials. *Mater Sci Eng B* **108**:143–147 (2004).
- 58 Poirier E, Chahine R, Benard P, Cossement D, Lafi L, Melancon E, *et al*, Storage of hydrogen on single-walled carbon nanotubes and other carbon structures. *Appl Phys A* **78**:961–967 (2004).
- 59 Bacsa R, Laurent C, Morishima R, Suzuki H and Le Lay M, Hydrogen storage in high surface area carbon nanotubes produced by catalytic chemical vapor deposition. *J Phys Chem B* **108**:12718–12723 (2004).
- 60 Ugarte D, Curling and closure of graphitic networks under electron–beam irradiation. *Nature* **359**:707–709 (1992).
- 61 Ugarte D, Onion-like graphitic particles. *Carbon* **33**:989–993 (1995).
- 62 Terrones H, Terrones M and Hsu WK, Beyond C_{60} : Graphite structures for the future. *Chem Soc Rev* **24**:341–350 (1995).
- 63 Saito Y, Yoshikawa T, Inagaki M, Tomita M and Hayashi T, Growth and structure of graphitic tubules and polyhedral particles in arc-discharge. *Chem Phys Lett* **204**:277–282 (1993).
- 64 Sano N, Wang H, Chhowalla M, Alexandrou I and Amarantunga GAJ, Synthesis of carbon 'onions' in water. *Nature* **414**:506–507 (2001).
- 65 de Heer WA and Ugarte D, Carbon onions produced by heat-treatment of carbon soot and their relation to the 2175 nm interstellar absorption feature. *Chem Phys Lett* **207**:480–486 (1993).
- 66 He C, Zhao N, Shi C, Du X, Li J and Cui L, A practical method for the production of hollow carbon onion particles. *J Alloys Comp* **425**:329–333 (2006).
- 67 Georgakilas V, Guldi DM, Signorini R, Bozio R and Prato M, Organic functionalization and optical properties of carbon onions. *J Am Chem Soc* **125**:14268–14269 (2003).
- 68 Rettenbacher AS, Elliott B, Hudson JS, Amirkhani A and Echegoyen L, Preparation and functionalization of multilayer fullerenes (carbon nanonions). *Chem Eur J* **12**:376–387 (2006).
- 69 Ugarte D, How to fill or empty a graphitic onion. *Chem Phys Lett* **209**:99–103 (1993).
- 70 Seshadri R, Sen R, Subbana GN, Kannan KR and Rao CNR, Iron, cobalt and nickel nanoparticles encapsulated in carbon obtained by the arc evaporation of graphite with metals. *Chem Phys Lett* **231**:308–313 (1994).
- 71 Jiao J, Seraphin S, Wang XK and Withers JC, Preparation and properties of ferromagnetic carbon-coated Fe, Co, and Ni nanoparticles. *J Appl Phys* **80**:103–108 (1996).
- 72 Sun L and Banhart F, Graphitic onions as reaction cells on the nanoscale. *Appl Phys Lett* **88**:1931211–3 (2006).
- 73 Viculis LM, Mack JJ and Kaner RB, A chemical route to carbon nanoscrolls. *Science* **299**:1361 (2003).
- 74 Shioyama H and Akita T, A new route to carbon nanotubes. *Carbon* **41**:179–181 (2003).
- 75 Bacon R, Growth, structure, and properties of graphite whiskers. *J Appl Phys* **31**:283–290 (1960).
- 76 Novoselov KS, Geim AK, Morozov SV, Jiang D, Zhang Y, Dubonos SV, *et al*, Electric field effect in atomically thin carbon films. *Science* **306**:666–669 (2004).
- 77 Zhang Y, Tan YW, Stormer HL and Kim P, Experimental observation of the quantum Hall effect and Berry's phase in graphene. *Nature* **438**:201–204 (2005).
- 78 Peres NMR, Guinea F and Castro Neto AH, Electronic properties of two-dimensional carbon. *Annals Phys* **321**:1559–1567 (2006).
- 79 Nomura K and MacDonald AH, Quantum Hall ferromagnetism in graphite. *Phys Rev Lett* **96**:2566021–4 (2006).
- 80 Rollings E, Gweon GH, Zhou SY, Mun BS, McChesney JL, Hussain BS, *et al*, Synthesis and characterization of atomically thin graphite films on silicon carbide substrate. *J Phys Chem Solids* **67**:2172–2177 (2006).
- 81 Mantell CL, *Carbon and Graphite Handbook*, Interscience Publishers, New York, Ch. 2 (1968).
- 82 Byrne JF and Marsh H, in *Porosity in Carbons*, ed by Patrick JW, Halsted Press, New York, Ch. 1 (1995).
- 83 Winter M and Besenhard JO, in *Lithium Ion Batteries – Fundamentals and Performance*, ed. by Wakihara M and Yamamoto O, Wiley–VCH, Weinheim, Germany, Ch. 6 (1998).
- 84 Mantell CL, *Carbon and Graphite Handbook*, Interscience Publishers, New York, Ch. 6 (1968).
- 85 Nijkamp MG, Raaymakers JEMJ, van Dillen AJ and de Jong KP, Hydrogen storage using physisorption – materials demands. *Appl Phys A* **72**:619–623 (2001).

- 86 Pierre AC and Pajonk GM, Chemistry of aerogels and their applications. *Chem Rev* **102**:4243–4265 (2002).
- 87 Al-Muhtaseb SA and Ritter JA, Preparation and properties of resorcinol–formaldehyde organic and carbon gels. *Adv Mater* **15**:101–114 (2003).
- 88 Reynolds GAM, Fung AWP, Wang ZH, Dresselhaus MS and Pekala RW, The effects of external conditions on the internal structure of carbon aerogels. *J Non-Cryst Solids* **188**:27–33 (1995).
- 89 Moreno–Castilla C and Maldonado–Hodar FJ, Carbon aerogels for catalysis applications: An overview. *Carbon* **43**:455–465 (2005).
- 90 Frackowiak E and Beguin F, Carbon materials for the electrochemical storage of energy in capacitors. *Carbon* **39**:937–950 (2001).
- 91 Robertson J, Amorphous carbon. *Curr Opin Sol State Mater Sci* **1**:557–561 (1996).
- 92 Lifshitz Y, Hydrogen-free amorphous carbon. *Diam Relat Mater* **5**:388–400 (1996).
- 93 Robertson J, Diamond-like carbon. *Pure Appl Chem* **66**:1789–1796 (1994).
- 94 Voevodin AA, Laube SJP, Walck SD, Solomon JS, Donley MS and Zabinski JSD, Pulsed laser deposition of diamond-like carbon. *J Appl Phys* **78**:4123–4130 (1995).
- 95 Weiler M, Sattel S, Giessen T, Jung K, Ehrhardt H, Veerasamy VS, *et al*, Preparation and properties of tetrahedral hydrogenated amorphous carbon. *Phys Rev B* **53**:1594–1608 (1996).
- 96 Rode AV, Hyde ST, Gamaly EG, Elliman RG, McKenzie DR and Bulcock S, Structural analysis of a carbon foam formed by high pulse-rate laser ablation. *Appl Phys A* **69**:S755–S758 (1999).
- 97 Rode AV, Elliman RG, Gamaly EG, Veinger AI, Christy AG, Hyde ST, *et al*, Electronic and magnetic properties of carbon nanofoam produced by high–repetition–rate laser ablation. *Appl Surf Sci* **197–198**:644–649 (2002).
- 98 Rode AV, Gamaly EG, Christy AG, Fitz Gerald JG, Hyde ST, Elliman RG, *et al*, Unconventional magnetism in all–carbon nanofoam. *Phys Rev B* **70**:054407-1–054407-9 (2004).
- 99 Gogotsi YG, Jeon ID and McNallan MJ, Carbon coatings on silicon carbide by reaction with chlorine-containing gases. *J Mater Chem* **7**:1841–1848 (1997).
- 100 Gogotsi Y, Nikitin A, Ye HH, Zhou W, Fischer JE, Yi B, *et al*, Nanoporous carbide-derived carbon with tunable pore size. *Nature Mater* **2**:591–594 (2003).
- 101 Dash RK, Nikitin A and Gogotsi Y, Microporous carbon derived from boron carbide. *Microp Mesop Mater* **72**:203–208 (2004).
- 102 Dash RK, Yushin G and Gogotsi Y, Synthesis, structure and porosity analysis of microporous mesoporous carbon derived from zirconium carbide. *Microp Mesop Mater* **86**:50–57 (2005).
- 103 Yushin GN, Hoffman EN, Nikitin A, Ye HH, Barsoum MW and Gogotsi Y, Synthesis of nanoporous carbide-derived carbon by chlorination of titanium silicon carbide. *Carbon* **43**:2075–2082 (2005).
- 104 Lee CY, Chiu HT, Peng CW, Yen MY, Chang YH and Liu CS, Polygon building block route to sp²-carbon-based materials. *Adv Mater* **13**:1105–1107 (2001).
- 105 Hu G, Ma D, Cheng M, Liu L and Bao X, Direct synthesis of uniform hollow carbon spheres by a self–assembly template approach. *Chem Commun* 1948–1949 (2002).
- 106 Chang YH, Chiu HT, Wang LS, Wan CY, Peng CW and Lee CY, Synthesis of sp² carbon nano- and microrods with novel structure and morphology. *J Mater Chem* **13**:981–982 (2003).
- 107 Gogotsi Y, Dash RK, Yushin G, Yildirim T, Laudisio G and Fischer JE, Tailoring of nanoscale porosity in carbide-derived carbons for hydrogen storage. *J Am Chem Soc* **127**:16006–16007 (2005).
- 108 Dash R, Chmiola J, Yushin G, Gogotsi Y, Laudisio G, Singer J, *et al*, Titanium carbide derived nanoporous carbon for energy-related applications. *Carbon* **44**:2489–2497 (2006).
- 109 Yushin G, Dash R, Jagiello J, Fischer JE and Gogotsi Y, Carbide-derived carbons: Effect of pore size on hydrogen uptake and heat of adsorption. *Adv Funct Mater* **16**:2288–2293 (2006).
- 110 Chmiola J, Yushin G, Dash RK, Hoffman EN, Fischer JE, Barsoum NW, *et al*, Double-layer capacitance of carbide derived carbons in sulfuric acid. *Electrochem Sol State Lett* **8**:A357–A360 (2005).
- 111 Yushin G, Hoffman EN, Barsoum NW, Gogotsi Y, Howell CA, Sandeman SR, *et al*, Mesoporous carbide-derived carbon with porosity tuned for efficient adsorption of cytokines. *Biomater* **27**:5755–5762 (2006).
- 112 Xiong Y, Xie Y, Li X and Li Z, Production of novel amorphous carbon nanostructures from ferrocene in low-temperature solution. *Carbon* **42**:1447–1453 (2004).
- 113 Xu L, Zhang W, Yang Q, Ding Y, Yu W and Qian Y, A novel route to hollow and solid carbon spheres. *Carbon* **43**:1090–1092 (2005).
- 114 Elings V and Wudl F, Tunneling microscopy on various carbon materials. *J Vac Sci Technol A* **6**:412–414 (1988).
- 115 Falcao EHL, Yeh Y, Dunn B and Wudl F, Electrochemical and physical chemical properties of sp² carbon microrods. *Carbon* **44**:1718–1724 (2006).
- 116 Falcao EHL, Yeh Y, Dunn BS and Wudl F, Carbon microrods as a high lithium capacity material (INOR–867). Presented at the 229th ACS National Meeting, San Diego, CA (2005).
- 117 Hart RW, White HS, Dunn B and Rolison DR, 3-D microbatteries. *Electrochem Commun* **5**:120–123 (2003).
- 118 Long JW, Dunn B, Rolison DR and White HS, Three-dimensional battery architectures. *Chem Rev* **104**:4463–4492 (2004).
- 119 Falcao EHL, Yeh Y, Min HS, Dunn BS and Wudl F, Use of carbon microrods in 3D electrodes (Y1–1230). Presented at the 208th Meeting of the Electrochemical Society, Los Angeles, CA (2005).